A PDF Approximation Formula for Amplitude of Rectangular Components of E-field in a Probabilistic Statistical Model of Reverberation Chamber

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***Abstract*—A more generalized probabilistic statistical model simulating the multipath fading electromagnetic environment in reverberation chamber has been proposed recently, which is superposition of the *M* plane waves with strong amplitude and the *N*−*M* plane waves with weak amplitude. But it has difficulty in obtaining a closed-form solution for the probability density function (PDF) of the amplitude of the rectangular component of the electric field. This paper proposes to approximate the theoretical PDF of the amplitude of the rectangular component of the electric field with the formula of the Rician distribution under the condition of *M* > 5, and the approximation formula is verified by MC simulations and data fitting.**

***Keywords*—Probability Density Function, Rician Distribution, Reverberation Chamber, Monte Carlo Simulation.**

I. INTRODUCTION

A reverberation chamber is an electrically large, multimode, high Q metal cavity, usually consisting of a metal shielded enclosure and a built-in stirrer. It can be used to simulate wireless communication channels and provide an environment for testing wireless communication devices [1].

Ref. [2], [3] modeled the angular spectrum stochastic plane-wave integrals of classical reverberation chambers and, combined with a number of assumptions about the stochasticity of the angular spectrum, derived probability density function (PDF) for a range of field. Ref. [4] showed that as the number of plane waves increases, the field statistics will be characterized more accurately. Ref. [5] discretized the plane wave integral expression in [2] to model plane wave superposition in a reverberation chamber and used Monte Carlo (MC) simulations to obtain probabilistic statistical field properties. Ref. [6] developed a probabilistic statistical model of the Rician distribution electromagnetic environment based on [2], [5] in conjunction with the physical characteristics of the Rician distribution. Ref. [7] established a more generalized probabilistic statistical model of the multipath fading electromagnetic environment in reverberation chamber.

This paper further discusses the PDF of the amplitude of the rectangular electric field component in the probabilistic statistical model proposed in [7]. Based on the derivation of the model under the Central Limit Theorem (CLT) and the characteristics of strong-weak superposition, the hypothesis of approximating the theoretical PDF of the amplitude of the rectangular component of the electric field with the formula of the Rician distribution is proposed and verified by MC simulations and data fitting.

II. PROBABILISTIC STATISTICAL MODEL

The probabilistic statistical model proposed by [7] is shown in (1):

$$\boldsymbol{E}(\boldsymbol{r}) = \frac{4\pi}{N}\left\{\sum_{m=1}^{M} \boldsymbol{F}_{\mathrm{s}}(\alpha_m,\beta_m)\exp[\mathrm{j}\boldsymbol{k}(\alpha_m,\beta_m)\cdot\boldsymbol{r}] + \sum_{n=M+1}^{N} \boldsymbol{F}_{\mathrm{w}}(\alpha_n,\beta_n)\exp[\mathrm{j}\boldsymbol{k}(\alpha_n,\beta_n)\cdot\boldsymbol{r}]\right\} \quad (1)$$

Equation (1) represents the electric field at the source-free region $\boldsymbol{r}$ in the reverberation chamber, where $m(m = 1,2,\ldots,M)$ cor-

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responds to the mth plane wave with strong electric field amplitude, $\boldsymbol{F}_{\mathrm{s}}(\alpha_m,\beta_m)$ is the corresponding electric field angular spectrum, $n(n=M+1,\ldots,N)$ corresponds to the $n-M$th plane wave with weak electric field amplitude, $\boldsymbol{F}_{\mathrm{w}}(\alpha_n,\beta_n)$ is the corresponding electric field angular spectrum. The angular spectrum of the electric field is a complex vector that varies with the rotation of the stirrer. $\alpha\in[0,\pi]$ is the angle of incidence, $\beta\in[0,2\pi]$ is the azimuthal angle in the interval. $\boldsymbol{k}$ is the vector wavenumber.

Considering $\boldsymbol{r}=\boldsymbol{0}$, (1) can be simplified as:

$$\boldsymbol{E}=\frac{4\pi}{N}\left\{\sum_{m=1}^{M}\boldsymbol{F}_{\mathrm{s}}(\alpha_m,\beta_m)+\sum_{n=M+1}^{N}\boldsymbol{F}_{\mathrm{w}}(\alpha_n,\beta_n)\right\} \tag{2}$$

where $\boldsymbol{F}_{\mathrm{s}}(\alpha_m,\beta_m)$ and $\boldsymbol{F}_{\mathrm{w}}(\alpha_n,\beta_n)$ can be expressed in components as:

$$\begin{aligned}\boldsymbol{F}_{\mathrm{s}}(\alpha_m,\beta_m)=&\hat{\boldsymbol{\alpha}}\left[F_{\mathrm{s}_\alpha\mathrm{r}}(\alpha_m,\beta_m)+\mathrm{j}F_{\mathrm{s}_\alpha\mathrm{i}}(\alpha_m,\beta_m)\right]\\&+\hat{\boldsymbol{\beta}}\left[F_{\mathrm{s}_\beta\mathrm{r}}(\alpha_m,\beta_m)+\mathrm{j}F_{\mathrm{s}_\beta\mathrm{i}}(\alpha_m,\beta_m)\right]\end{aligned} \tag{3}$$

$$\begin{aligned}\boldsymbol{F}_{\mathrm{w}}(\alpha_n,\beta_n)=&\hat{\boldsymbol{\alpha}}\left[F_{\mathrm{w}_\alpha\mathrm{r}}(\alpha_n,\beta_n)+\mathrm{j}F_{\mathrm{w}_\alpha\mathrm{i}}(\alpha_n,\beta_n)\right]\\&+\hat{\boldsymbol{\beta}}\left[F_{\mathrm{w}_\beta\mathrm{r}}(\alpha_n,\beta_n)+\mathrm{j}F_{\mathrm{w}_\beta\mathrm{i}}(\alpha_n,\beta_n)\right]\end{aligned} \tag{4}$$

where subscript r and i represent the real and imaginary parts, $\hat{\boldsymbol{\alpha}}$ and $\hat{\boldsymbol{\beta}}$ are the unit vectors in the α and β directions. In (3), $F_{\mathrm{s}_\alpha\mathrm{r}}(\alpha_m,\beta_m)$ and $F_{\mathrm{s}_\beta\mathrm{r}}(\alpha_m,\beta_m)$ are assumed to be $A_m\cos\theta_m$, $F_{\mathrm{s}_\alpha\mathrm{i}}(\alpha_m,\beta_m)$ and $F_{\mathrm{s}_\beta\mathrm{i}}(\alpha_m,\beta_m)$ are assumed to be $A_m\sin\theta_m$. A_m is the parameter determining the amplitude of the mth plane wave with strong electric field amplitude. Assuming the corresponding phase θ_m follows the uniform distribution $U[0,2\pi)$. In (4), Assuming $F_{\mathrm{w}_\alpha\mathrm{r}}(\alpha_n,\beta_n)$, $F_{\mathrm{w}_\alpha\mathrm{i}}(\alpha_n,\beta_n)$, $F_{\mathrm{w}_\beta\mathrm{r}}(\alpha_n,\beta_n)$ and $F_{\mathrm{w}_\beta\mathrm{i}}(\alpha_n,\beta_n)$ follow the normal distribution $N(0,{\sigma_n}^2)$.

According to (3) and (4), the x-component of the electric field E_x in (2) can be expressed as the superposition of the strong component electric field $E_{x\mathrm{s}}$ and the weak component electric field $E_{x\mathrm{w}}$:

$$\begin{aligned}E_x&=E_{x\mathrm{s}}+E_{x\mathrm{w}}=E_{x\mathrm{s_r}}+\mathrm{j}E_{x\mathrm{s_i}}+E_{x\mathrm{w_r}}+\mathrm{j}E_{x\mathrm{w_i}}\\&=\sum_{m=1}^{M}E_{x\mathrm{s}_m}\exp(\mathrm{j}\theta_m)+E_{x\mathrm{w}}\end{aligned} \tag{5}$$

where each term in (5) is uncorrelated and $E_{x\mathrm{s}_m}$ is determined by A_m. Assuming $E_{x\mathrm{w_r}}$ and $E_{x\mathrm{w_i}}$ follow the normal distribution $N(0,{\sigma_x}^2)$.

III. Inference of Probability Density Function

Using the method of characteristic functions to derive the PDF, and employing [8], the general solution for the PDF of amplitude $|E_x|$ is obtained for any given value of M, as shown in (6):

$$f_R(|E_x|)=\begin{cases}\dfrac{|E_x|}{{\sigma_x}^2}\exp\left(\dfrac{-|E_x|^2}{2{\sigma_x}^2}\right), M=0\\ \dfrac{|E_x|}{{\sigma_x}^2\pi^{M-1}}\displaystyle\int_0^{\pi}\cdots\int_0^{\pi}\int_0^{\pi}\exp\left(-\frac{|E_x|^2+t_M}{2{\sigma_x}^2}\right)\\ \cdot I_0\left(\dfrac{|E_x|\sqrt{t_M}}{{\sigma_x}^2}\right)d\theta_1\,d\theta_2\ldots d\theta_{M-1}, M>0\end{cases} \tag{6}$$

where t_M follows the recursive relationship:

$$t_M=\begin{cases}0, & M=0\\ \displaystyle\sum_{m=1}^{M}E_{x\mathrm{s}_m}^2-\sum_{m=1}^{M}2E_{x\mathrm{s}_m}\sqrt{t_{m-1}}\cos\theta_{m-1}, & M>0\end{cases} \tag{7}$$

Due to the presence of multiple integrals in (6), a closed-form solution has not been obtained. The computation of (6) becomes challenging, particularly when M is large.

A. *Proposed Hypothesis*

According to the CLT, when the value of M is sufficiently large, both the real and imaginary parts of E_x follow the normal distribution $N\left(0,\frac{1}{2}\sum_{m=1}^{M}E_{x\mathrm{s}_m}^2+{\sigma_x}^2\right)$, the amplitude $|E_x|$ follows a Rayleigh distribution:

$$f_R(|E_x|)=\frac{|E_x|}{{\sigma^2}_{\mathrm{rayleigh}}}\exp\left(\frac{-|E_x|^2}{2{\sigma^2}_{\mathrm{rayleigh}}}\right) \tag{8}$$

where

$${\sigma^2}_{\mathrm{rayleigh}}=\frac{1}{2}\sum_{m=1}^{M}E_{x\mathrm{s}_m}^2+{\sigma_x}^2 \tag{9}$$

The Rayleigh distribution is a special case of the Rician distribution. Additionally, due to the similarity between the characteristics of the strong-weak superposition in this model and the physical characteristics of the Rician distribution electromagnetic environment, therefore, it is hypothesized that as M increases sufficiently, the PDF of the amplitude $|E_x|$ can be approximated by the expression of the Rician distribution in (10).

$$f_R(|E_x|)=\frac{|E_x|}{{\sigma^2}_{\mathrm{rician}}}\exp\left(-\frac{|E_x|^2+s^2}{2{\sigma^2}_{\mathrm{rician}}}\right)I_0\left(\frac{|E_x|s}{{\sigma^2}_{\mathrm{rician}}}\right) \tag{10}$$

Further, we assume that there is a linear relationship between ${\sigma^2}_{\mathrm{rician}}$ in (10) and ${\sigma^2}_{\mathrm{rayleigh}}$ in (9), as well as s^2 and $\sum_{m=1}^{M}{E_{x\mathrm{s}_m}}^2$, as shown in (11),

$$\begin{cases}{\sigma^2}_{\mathrm{rician}}=k_1{\sigma^2}_{\mathrm{rayleigh}}+b_1\\ s^2=k_2\displaystyle\sum_{m=1}^{M}E_{x\mathrm{s}_m}^2+b_2\end{cases}, \tag{11}$$

where k_1, k_2, b_1 and b_2 are undetermined coefficients.

As M increases sufficiently, k_1 will increase with the growth of M and gradually approach 1, k_2 will decrease with the growth of M and gradually approach 0, and b_1 and b_2 should also approach 0.

B. *MC Simulations and Data Fitting*

To verify the feasibility of (10) and (11), MC simulations are conducted on (2) with a sample size of 5000 and $N=1000$, under various scenarios for the sum of squared amplitudes $\sum_{m=1}^{M}E_{x\mathrm{s}_m}^2$. Using a PDF fitting tool, the data from the MC simulations are fitted with the PDF shown in (10), simultaneously record the different results of the two parameters, ${\sigma^2}_{\mathrm{rician}}$ and s^2, obtained from each fitting, and verified the linear relationship shown in (11).

To determine the coefficients in (11), let M take values of 5, 6, 7, 8, 10, 12, 15, 20, 25, 30, 40, 50. Under each condition of M, plot the relationship between ${\sigma^2}_{\mathrm{rician}}$ and corresponding ${\sigma^2}_{\mathrm{rayleigh}}$, as well as the relationship between s^2 and $\sum_{m=1}^{M}{E_{x\mathrm{s}_m}}^2$, and obtain the fitting values of k_1, k_2, b_1 and b_2 through a linear fitting tool, as shown in Table I.

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TABLE I
THE FITTED VALUES OF COEFFICIENTS FOR DIFFERENT VALUES OF M

M	k_1	k_2	b_1	b_2
5	0.5494	0.448	0.1407	-0.1874
6	0.5904	0.4068	0.09299	-0.07493
7	0.6276	0.3814	-0.1674	0.2673
8	0.6287	0.3673	0.379	-0.1589
10	0.6418	0.3588	-0.147	0.4663
12	0.6669	0.3374	-0.5155	1.011
15	0.6636	0.3388	0.04229	-0.01523
20	0.6997	0.3024	0.3354	-0.4364
25	0.7363	0.2592	0.07228	-0.1519
30	0.7481	0.253	-0.03387	-0.1036
40	0.7447	0.2501	0.8886	-1.546
50	0.7679	0.2247	-0.6142	1.516

It can be observed that k_1 increases with the growth of M, while k_2 decreases with the growth of M from Table I, consistent with the anticipated trend. Moreover, it is found that the sum of k_1 and k_2 under different conditions is approximately equal to 1. Both b_1 and b_2 do not exhibit a clear trend, and many data points tend to be close to 0. After setting b_1 and b_2 to 0, MC simulation is conducted on the original case where b_1 and b_2 were relatively large ($M = 12$). The result as shown in Fig. 1 indicates that the Rician distribution in (10) still closely approximates the PDF of $|E_x|$ in (6), and the approximation is even better than before. Therefore, b_1 and b_2 in (11) can be 0.

The power function is selected to fit the functional relationship between k_2 and M, and the fitting result is shown in (12). The calculated sum of squared errors (SSE) is 0.001493, the root mean square error (RMSE) is 0.01222, the coefficient of determination (R-square) is 0.9723, and the adjusted R-square is 0.9695, indicating a good fit.

$$k_2 = 0.6824M^{-0.2822}, M > 5 \tag{12}$$

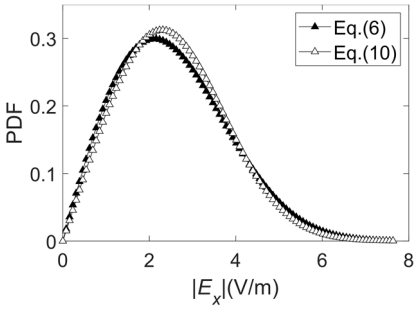
Since the sum of k_1 and k_2 is approximately equal to 1, k_1 can be expressed as $1 - k_2$. The functional relationship between k_1 and M is shown in (13). The calculated SSE is 0.001730, the RMSE is 0.012006, and the R-square is 0.967043, indicating a similarly good fit.

$$k_1 = 1 - 0.6824M^{-0.2822}, M > 5 \tag{13}$$

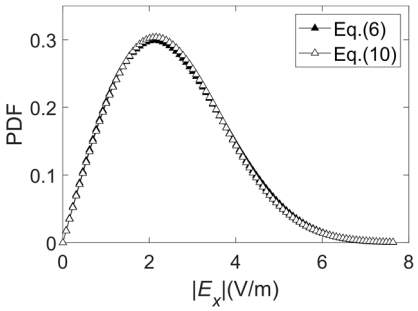
Further functional relationships between ${\sigma^2}_{\text{rician}}$ and M, as well as between s^2 and M, can be obtained from (12) and (13). This determined the specific expressions of (10) under different conditions for $M > 5$, obtaining a closed-form approximate solution.

IV. CONCLUSION

Based on the model proposed by [7], the PDF of the amplitude $|E_x|$ is derived under the CLT, and considering the characteristics of this model, this paper proposes to approximate the theoretical PDF of $|E_x|$ with the formula of the Rician distribution under the condition of $M > 5$, and the approximation formula is verified by MC simulations and data fitting.



(a) M=12, b_1=−0.5155, b_2=1.011



(b) M=12, b_1=0, b_2=0

Figure 1. Effect of b_1 and b_2 taking the value of 0 in (10)

Assuming the forms of the two parameters, ${\sigma^2}_{\text{rician}}$ and s^2, in (10). Through extensive data fitting, the expression relationship between these two parameters and M is determined, thereby specifying the specific form of the Rician distribution in (10). This allows for the substitution of (10) for (6) when $M > 5$, providing a closed-form approximation and addressing the computational complexity associated with multiple integrals in (6) when the order is large.

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